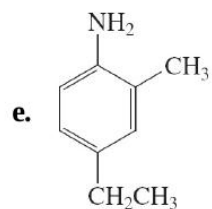
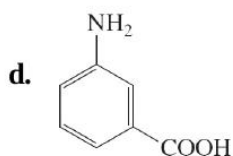
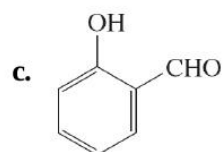
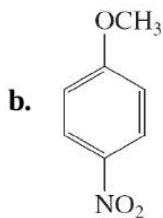
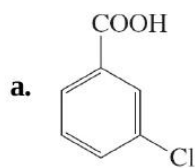


36. Name each of the following compounds by using the IUPAC system and, if possible, a reasonable common alternative. (**Hint:** The order of functional group precedence is $\text{-COOH} > \text{-CHO} > \text{-OH} > \text{-NH}_2$.)

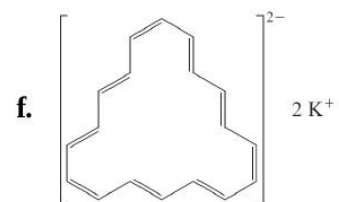
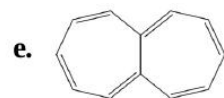
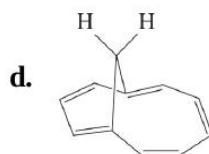
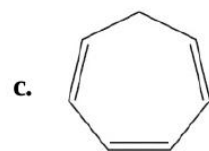
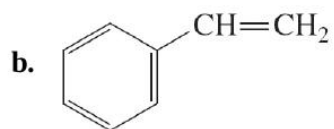


38. Draw the structure of each of the following compounds. If the name itself is incorrect, give a correct systematic alternative. **(a)** *o*-Chlorobenzaldehyde; **(b)** 2,4,6-trihydroxybenzene; **(c)** 4-nitro-*o*-xylene; **(d)** *m*-isopropylbenzoic acid; **(e)** 4,5-dibromoaniline; **(f)** *p*-methoxy-*m*-nitroacetophenone.

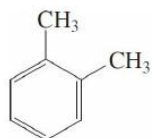
39. The complete combustion of benzene is exothermic by approximately $-789 \text{ kcal mol}^{-1}$. What would this value be if benzene lacked aromatic stabilization?

41. Complete hydrogenation of 1,3,5,7-cyclooctatetraene is exothermic by

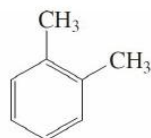
42. Which of the following structures qualify as being aromatic, according to Hückel's rule?



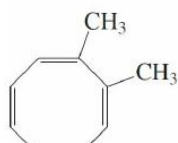
58. Because of cyclic delocalization, structures A and B shown here for *o*-dimethylbenzene (*o*-xylene) are simply two resonance forms of the same molecule. Can the same be said for the two dimethylcyclooctatetraene structures C and D? Explain.



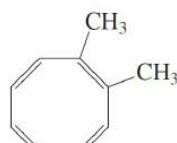
A



B



C



D

- 66.** Explain the following reaction and the indicated stereochemical result mechanistically.